

Hugo JARKOFF

Machine Learning Software Engineer (MSc.)

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- in/hugojarkoff
- Fontainebleau, France

ML Software Engineer with 5+ years of experience building and deploying ML systems at scale.

EXPERIENCE

- **Addactis (formerly NamR, acquired in 2025)**

ML/CV Software Engineer

Responsible for production ML systems powering aerial imagery processing and SaaS products:

- Refactored the team’s ML training framework, reducing time from data acquisition to production release by over 50%.
 - Redesigned the aerial imagery preprocessing pipeline with serverless Cloud Run APIs, cutting legacy codebase size down and reducing data integration time for ML workflows.
 - Refactored the aerial imagery inference pipeline into a distributed GPU batch-processing system with PostgreSQL-coordinated workers, achieving an 80% faster runtime and 90% lower compute cost on large-scale datasets.
 - Designed, built, and deployed an AI agent platform (FastAPI + Pydantic-AI) integrated into production SaaS insurance products.

Paris, France
2024 - present
- **Finegrain**

ML/CV Software Engineer

State-of-the-art generative model fine-tuning for image editing:

- Fine-tuned foundation models (SAM, Stable Diffusion) using adapters (LoRA, IP-Adapter) for image editing tasks.
 - Conducted continuous literature review in generative modeling.
 - Contributed to the team’s high-performance OSS micro-framework for foundation model adaptation (Refiners).

Paris, France
2024 (4 months)
- **Invoxia**

R&D Machine Learning Engineer

Led multiple end-to-end ML projects from research to deployment:

- Designed a CV-based signal processing model to estimate canine respiratory rate from accelerometer data, achieving 98.5% accuracy (preprint available).
 - Researched and adapted neural architectures (CNNs, Vision Transformers), focusing on model compression for embedded deployment.
 - Contributed to the inference system for trained models, serving 10k+ devices via batch processing through Kubernetes and AWS SageMaker, with real-time monitoring platform (Grafana, Prometheus).

Paris, France
2021 – 2024

EDUCATION

- **ISAE-SUPAERO**

Engineering Degree (MSc.)

- *Major in Machine Learning:* Deep Learning (Computer Vision, NLP), RL, MLOps.
 - *Minor in Advanced Mathematics:* Bayesian statistics, HPC, optimization.
 - *Final year Team Project for Airbus:* Developed an airport image segmentation and classification system using fully convolutional neural networks (UNet).

Toulouse, France
2016 – 2020
- **Lycée Louis-le-Grand**

Preparatory Classes

- *PCSI - PC**: Mathematics - Physics - Chemistry - Computer Science.

Paris, France
2014 – 2016

TECHNICAL SKILLS

- **Programming:** advanced knowledge of Python (multithreaded programming, async programming); intermediate knowledge of C++ / C / Bash.
- **Deep Learning:** PyTorch, TensorFlow, Keras.
- **MLOps & Infrastructure:** Docker, Kubernetes, Terraform, NVIDIA Triton, PyTriton, CI/CD (GitLab), experiment tracking (WandB, ClearML).
- **Databases:** PostgreSQL, PostGIS.
- **Cloud Platforms:** Google Cloud Platform (GCP), Amazon Web Services (AWS).

LANGUAGES

- **French:** Native language.
- **English:** Full professional proficiency.
TOEFL ITP: 633/677.

PERSONAL PROJECTS AND DIVERSE INTERESTS

- **Language Models Training:** RapGPT: Trained a Transformer-based language model from scratch (architecture, training loop, and deployment) to generate French rap lyrics; deployed live demos on Hugging Face (with open weights).
- **Systems Programming:** Developed a simple, lightweight (700KB running), multiprocess (fork) HTTP server in C++.
- **Rock Climbing:** Experienced climber (20+ years of practice). Designed and built a custom finger-force measurement device (ESP32 soldered with deformation gauge, embedded with C software).